- To connect your VPCs to your own network, see Site-to-Site VPN scenarios in the *AWS Site-to-Site VPN User Guide*.
- To connect your VPCs to each other and to your own network, see Example transit gateway scenarios in the *Amazon VPC Transit Gateways*.

Additional resources

- Understand resiliency patterns and trade-offs (AWS Architecture Blog)
- Plan your network topology (AWS Well-Architected Framework)
- Amazon Virtual Private Cloud Connectivity Options (AWS Whitepapers)

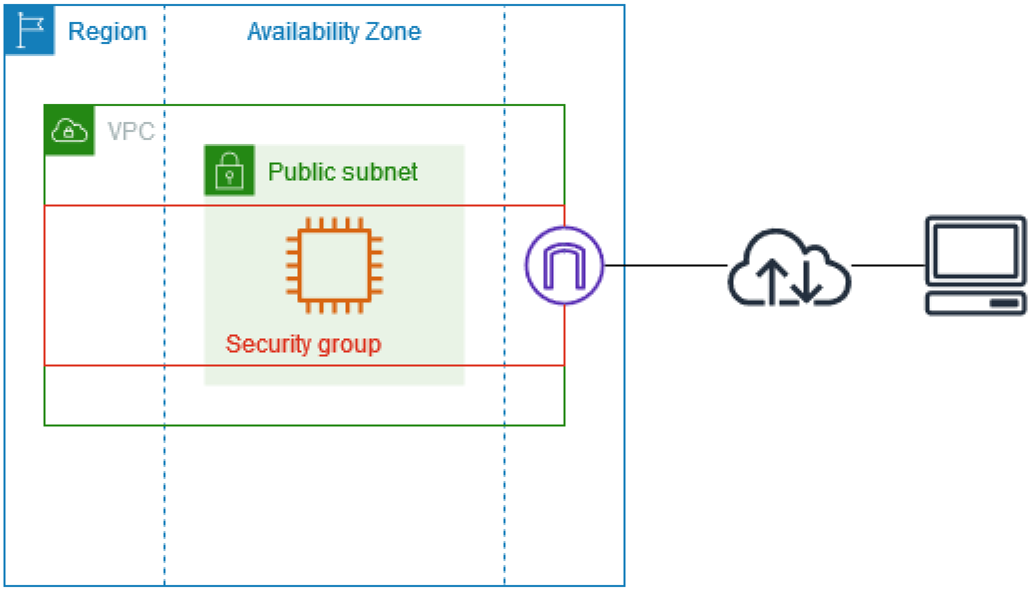
Example: VPC for a test environment

This example demonstrates how to create a VPC that you can use as a development or test environment. Because this VPC is not intended to be used in production, it is not necessary to deploy your servers in multiple Availability Zones. To keep the cost and complexity low, you can deploy your servers in a single Availability Zone.

Contents

Overview

The following diagram provides an overview of the resources included in this example. The VPC has a public subnet in a single Availability Zone and an internet gateway. The server is an EC2 instance that runs in the public subnet. The security group for the instance allows SSH traffic from your own computer, plus any other traffic specifically required for your development or testing activities.



Routing

When you create this VPC by using the Amazon VPC console, we create a route table for the public subnet with local routes and routes to the internet gateway. The following is an example of the route table with routes for both IPv4 and IPv6. If you create an IPv4-only subnet instead of a dual stack subnet, your route table has only the IPv4 routes.

Destination	Target
10.0.0.0/16	local
2001:db8:1234:1a00::/56	local
0.0.0.0/0	*igw-id*
::/0	*igw-id*

Security

For this example configuration, you must create a security group for your instance that allows the traffic that your application needs. For example, you might need to add a rule that allows SSH traffic from your computer or HTTP traffic from your network.

The following are example inbound rules for a security group, with rules for both IPv4 and IPv6. If you create IPv4-only subnets instead of dual stack subnets, you need only the rules for IPv4.

Source	Protocol	Port range	Description
0.0.0.0/0	TCP	80	Allows inbound HTTP access from all IPv4 addresses
::/0	TCP	80	Allows inbound HTTP access from all IPv6 addresses
0.0.0.0/0	TCP	443	Allows inbound HTTPS access from all IPv4 addresses
::/0	TCP	443	Allows inbound HTTPS access from all IPv6 addresses
Public IPv4 address range of your network	TCP	22	(Optional) Allows inbound SSH access from IPv4 IP addresses in your network
IPv6 address range of your network	TCP	22	(Optional) Allows inbound SSH access from IPv6 IP addresses in your network
Public IPv4 address range of your network	TCP	3389	(Optional) Allows inbound RDP access from IPv4 IP addresses in your network
IPv6 address range of your network	TCP	3389	(Optional) Allows inbound RDP access from IPv6 IP addresses in your network

1. Create the VPC

Use the following procedure to create a VPC with a public subnet in one Availability Zone. This configuration is suitable for a development or testing environment.

To create the VPC

1. Open the Amazon VPC console at https://console.aws.amazon.com/vpc/.

2. On the dashboard, choose **Create VPC**.
3. For **Resources to create**, choose **VPC and more**.
4. **Configure the VPC**

    a. For **Name tag auto-generation**, enter a name for the VPC.

    b. For **IPv4 CIDR block**, you can keep the default suggestion, or alternatively you can enter the CIDR block required by your application or network. For more information, see the section called "VPC CIDR blocks".

    c. (Optional) If your application communicates by using IPv6 addresses, choose **IPv6 CIDR block**, **Amazon-provided IPv6 CIDR block**.

5. **Configure the subnets**

    a. For **Number of Availability Zones**, choose **1**. You can keep the default Availability Zone, or alternatively you can expand **Customize AZs** and select an Availability Zone.

    b. For **Number of public subnets**, choose **1**.

    c. For **Number of private subnets**, choose **0**.

    d. You can keep the default CIDR block for the public subnet, or alternatively you can expand **Customize subnet CIDR blocks** and enter a CIDR block. For more information, see the section called "Subnet CIDR blocks".

6. For **NAT gateways**, keep the default value, **None**.
7. For **VPC endpoints**, choose **None**. A gateway VPC endpoint for S3 is used only to access Amazon S3 from private subnets.
8. For **DNS options**, keep both options selected. As a result, your instance will receive a public DNS hostname that corresponds to its public IP address.
9. Choose **Create VPC**.

2. Deploy your application

There are a variety of ways to deploy EC2 instances. For example:

- Amazon EC2 launch instance wizard
- Amazon EC2 Auto Scaling
- AWS CloudFormation
- Amazon Elastic Container Service (Amazon ECS)